

Travel Planner and Booking App

A Advance Java Project Report Submitted

In

COMPUTER SCIENCE & ENGINEERING

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DECLARATION

We hereby declare that the work presented in this report entitled “**Travel Planner and Booking App**”, was carried out by us. We have not submitted the matter embodied in this report for the award of any other degree or diploma of any other University or Institute. We have given due credit to the original authors/sources for all the words, ideas, diagrams, graphics, computer programs, experiments, results, that are not my original contribution. We have used quotation marks to identify verbatim sentences and given credit to the original authors. We affirm that no portion of our work is plagiarized, and the experiments and results reported in the report are not manipulated. In the event of a complaint of plagiarism and the manipulation of the experiments and results, we shall be fully responsible and answerable.

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Certified that **Keshab Kumar** (Roll No_1: 2201330100132) and **Avarnita Srivastava** (Roll No_2: 2201330100078) have carried out the research work presented in this Project Report entitled “**Travel Planner and Booking App**” for the award of **Bachelor of Technology, Computer Science & Engineering** from Dr. APJ Abdul Kalam Technical University, Lucknow under our supervision. The Project Report embodies results of original work, and studies are carried out by the students herself/himself. The contents of the Project Report do not form the basis for the award of any other degree to the candidate or to anybody else from this or any other University/Institution.

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ABSTRACT

The **Smart Travel Planner and Booking System** is a comprehensive, web-based application designed to streamline the entire process of trip planning, booking, and management through a centralized and intelligent platform. Built using **Advanced Java technologies** including **Java Servlets, JSP, Spring Framework, JDBC**, and integrated with modern APIs and AI features, this project aims to provide a seamless, personalized, and efficient travel experience for users and administrators alike.

Traditionally, travel planning involves navigating through multiple fragmented platforms to search destinations, book transport and accommodations, and manage itineraries—leading to a disjointed and time-consuming process. Our system addresses this issue by offering a unified portal that enables users to **register, explore travel destinations, plan itineraries, and book flights, trains, buses, and activities** all in one place. The application further incorporates **secure OTP-based password recovery, real-time data integration via external travel APIs, and payment gateway support** for reliable transactions.

What sets this project apart is the integration of **unique and intelligent features** that enhance user experience and system capabilities:

- An **AI-powered Itinerary Generator** recommends personalized travel plans based on user preferences, interests, and budget, using machine learning techniques.
- A **Smart Travel Chatbot**, powered by natural language processing, offers real-time assistance, answering travel-related queries and guiding users through bookings.
- **Location-based suggestions** leverage geolocation data to recommend nearby attractions and services, enriching the travel planning experience.
- A **Weather-based Packing List Generator** uses weather APIs to dynamically suggest packing essentials for the selected destination and date.
- A **Carbon Footprint Calculator** estimates the environmental impact of chosen travel options and promotes eco-conscious decisions.
- An **Admin Analytics Dashboard** visualizes real-time data on bookings, revenues, and user trends through interactive charts and reports.

The backend architecture follows the **MVC design pattern**, ensuring modularity, maintainability, and scalability. The system is supported by a **MySQL relational database**, and all features are accessible through a responsive web interface built using JSP, HTML5, CSS3, and JavaScript. The platform is deployable on cloud environments such as AWS or Heroku and can be extended with CI/CD pipelines for continuous deployment.

Through this project, we demonstrate a scalable and innovative solution tailored for modern travelers and travel agencies. It combines **advanced Java programming, API integration, and AI-powered user personalization** into a single, user-friendly platform—positioning it as a forward-thinking system for the evolving travel industry.

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CHAPTER 1

INTRODUCTION

1.1 BACKGROUND:

Travel has become an integral part of modern life, be it for leisure, business, education, or emergency needs. However, the process of planning a trip—from searching destinations to booking flights, accommodations, and local transportation—can be overwhelming and time-consuming, especially when users are forced to rely on multiple disjointed platforms. This fragmented approach leads to redundant data input, uncoordinated planning, and poor user experience.

Traditional travel booking platforms primarily focus on isolated services, such as airline or hotel booking, without offering integrated tools for complete trip management. They also lack intelligent features like personalized recommendations, dynamic itineraries, and real-time interaction. Users often find themselves switching between multiple apps or websites to accomplish what should be a simple, unified task.

This project, Smart Travel Planner and Booking System, addresses these issues by offering a centralized, intelligent web application that automates and personalizes the travel planning process. It is developed using Advanced Java technologies—including Java Servlets, JSP, JDBC, and the Spring Framework—and enhanced with external APIs and AI-driven features.

The application not only allows users to register, log in, and book various travel services but also adds a layer of smart decision-making through features such as:

AI-based itinerary generation,

location-aware recommendations,

weather-integrated packing suggestions, and

eco-awareness through carbon footprint estimations.

In addition, it incorporates a secure admin dashboard for monitoring system activities, managing bookings, analyzing user trends, and generating reports.

By addressing both user and administrative needs, the system presents a next-generation solution for travel management—efficient, user-friendly, and intelligent.

1.2 KEY FEATURES:

The Smart Travel Planner and Booking App offers a wide range of traditional and advanced features.

These include:

Basic Features

- **User Registration and Login** with secure password handling
- **Forgot Password** functionality with **OTP email verification**
- **Search and Book:**
 - Flights
 - Trains
 - Buses
- **Destination Search** and filtering by category

API Integration

- Integration with **travel APIs** (Amadeus or Skyscanner) for real-time destination, flight, and hotel data
- Integration with **weather API** (OpenWeatherMap) to check destination conditions
- Integration with **payment gateways** (Stripe, PayPal) for secure bookings

Advanced Features

- **AI-based Itinerary Generator:** Recommends personalized trip plans based on user history, interests, and budget.
- **Smart Chatbot Assistant:** Helps users find destinations, answer FAQs, and guide bookings in real time.
- **Location-Based Suggestions:** Shows nearby tourist spots and hotels using geolocation data.
- **Smart Packing List:** Suggests what to pack based on travel dates and expected weather conditions.
- **Carbon Footprint Calculator:** Estimates environmental impact of selected transportation modes.
- **Admin Analytics Dashboard:** Visualizes real-time data on user bookings, revenue, and trends via graphs and charts.

Technologies Used

- Backend: Java Servlets, Spring MVC, JDBC
- Frontend: JSP, HTML5, CSS3, JavaScript, Bootstrap
- Database: MySQL
- Deployment: Apache Tomcat / Spring Boot
- Tools: Eclipse/IntelliJ, Maven, GitHub, Selenium, JUnit

1.3 Objective And Scope:

The primary objective of this project is to build a **smart, centralized, and user-centric travel planning system** that:

- Minimizes the effort and complexity involved in planning trips.
- Provides real-time booking and personalized suggestions.
- Combines multiple services—flights, trains, buses, and accommodations—into a single interface.
- Enhances the user experience using AI and automation.
- Ensures scalability and maintainability using a modular Java architecture.

The system is designed to serve:

- **General users:** To plan, book, and manage personal or group travel with ease.
- **Travel agencies or startups:** To adopt the platform as a customizable solution.
- **Admins:** To track system performance, user behavior, and booking trends.

The application can be extended further to include:

- Hotel room selection with room photos and filters
- Real-time maps and route visualizations
- In-app social features (e.g., travel forums, sharing itineraries)
- Mobile application support for Android/iOS

Overall, the system aims to become a **one-stop, intelligent travel planning solution** that evolves with user needs and industry trends.

CHAPTER 2

LITERATURE REVIEW:

2.1 Overview

Travel planning is an essential component of modern lifestyle—whether for leisure, work, or emergencies. Over the past two decades, the digital transformation of this domain has led to a surge in travel-based applications and booking platforms. While there has been significant progress in online ticketing systems, destination search engines, and hospitality booking services, these solutions remain highly fragmented and lack centralized user experiences.

The literature on travel information systems broadly categorizes platforms into siloed functionalities—flight booking, hotel reservations, tourist guides, or travel blogs. Rarely does a single system provide end-to-end coverage of the travel planning journey, from exploration to payment. Furthermore, these systems often ignore user preferences, environmental concerns, and real-time adaptability.

In recent years, emerging technologies such as Artificial Intelligence (AI), Natural Language Processing (NLP), and geo-intelligence have shown promise in transforming how users plan and experience travel. Advanced research is now focusing on building intelligent travel recommender systems, chatbot-based assistants, and itinerary generators. These systems aim to bring context-awareness and personalization into the travel lifecycle.

This project explores and leverages findings from existing systems, research publications, and market trends to design a unified, AI-powered Smart Travel Planner and Booking App. The purpose of this review is to understand traditional methods, their limitations, and how contemporary research and innovations can inform a more advanced and efficient solution.

2.2 Traditional methods :

Historically, travel planning was a manual, time-consuming, and heavily paper-based process. Individuals or families had to visit multiple service providers such as travel agents, airlines, train stations, or tourism offices. They relied on guidebooks, physical maps, and word-of-mouth suggestions to create itineraries. Even with the advent of telephonic reservations and early web-based tools, most users lacked access to unified data sources, making it difficult to compare options and make informed decisions.

In the early 2000s, the growth of online platforms like MakeMyTrip, Expedia, and IRCTC revolutionized how users booked travel. These platforms enabled basic self-service capabilities—users could check prices, book tickets, and access limited information about destinations. However, each platform generally focused on a single service (e.g., just flights or just trains), forcing users to switch between platforms for accommodation, transport, and activities.

Moreover, these traditional digital systems operated without much personalization. Users were shown generic suggestions, regardless of their preferences, budget, location, or history. There was no smart itinerary generation, no packing list recommendations, and minimal security layers for payment handling. Admin functions were also rudimentary, focused primarily on transaction logs without visual dashboards or analytics.

In conclusion, traditional methods—both manual and early digital—offered the basic building blocks for online travel management but lacked the intelligence, adaptability, and unified experience demanded by modern travelers.

2.3 Challenges of traditional method :

Despite their initial convenience, traditional travel planning methods—both manual and early-stage digital systems—suffered from several challenges that affected both users and administrators. These challenges are multifaceted, spanning technical, operational, and user-experience domains.

1. Fragmentation of Services

Users had to navigate multiple platforms (flight, hotel, bus, train, taxi, activities) to complete a single itinerary. This increased cognitive load, time consumption, and the likelihood of making errors.

2. Lack of Personalization

Traditional platforms did not maintain detailed user profiles or history, resulting in non-contextual and irrelevant suggestions. Travelers had to manually filter data without any intelligent guidance.

3. Static and Generic Itineraries

Most systems lacked dynamic itinerary builders. Users were responsible for manually assembling plans across websites and emails. There was no AI or logic to suggest optimal routes, timings, or bundling options.

4. No Real-Time Context

Weather conditions, local events, transportation delays, and user location were not considered during planning. Systems worked on static information models that could not adapt.

5. Poor Admin Tools

Admin panels were basic or nonexistent. There were limited options for tracking revenue, user activity, popular destinations, or customer support tickets.

6. Security & Privacy Risks

Early platforms stored user and payment data with minimal encryption. OTP-based verification, secure payment APIs, or GDPR compliance were not prioritized.

7. No Integration of AI or Automation

Tasks like packing lists, eco-travel suggestions, or FAQ resolution were manual or absent. There was no chatbot support, AI itinerary generation, or carbon footprint awareness.

Addressing these challenges requires a shift from transactional systems to intelligent, unified, and real-time platforms—precisely what this project aims to deliver.

2.4 Information In Various Research Papers :

Several research papers and academic studies have explored the application of advanced technologies in travel planning and intelligent information systems. The following highlights key findings from selected studies relevant to this project:

[1] An Intelligent Travel Recommendation System Using Collaborative Filtering– IEEE Xplore

Summary: This paper proposes a travel recommender system using collaborative filtering to suggest destinations based on user preferences and previous behavior.

Relevance: Inspired our use of ML models to generate AI-based personalized itineraries.

[2] Design and Implementation of a Chatbot System for Hotel Booking – International Journal of AI & Applications

Summary: Presents a chatbot using NLP to automate the hotel reservation process.

Relevance: Informs the chatbot module we integrated for real-time support and user guidance.

[3] Integrating Weather Forecast and Travel Planning Systems – SpringerLink

Summary: The study proposes using live weather data to influence travel planning decisions.

Relevance: Validates our packing list and smart recommendation feature that adapts based on weather.

[4] Carbon Footprint Estimation for Transport Modes – Journal of Environmental Management

Summary: Presents formulas and methods to calculate carbon emissions based on distance and transport mode.

Relevance: Enabled our implementation of a carbon footprint estimator to promote eco-conscious travel.

[5] A Survey on Tourism Information Systems – ACM Digital Library

Summary: Discusses the architecture and modules used in modern tourism platforms.

Relevance: Helped in defining our modular approach (user, admin, booking, analytics, etc.).

[6] Geolocation-Based Recommendation Systems in Tourism – Elsevier

Summary: Describes how user location can enhance context-aware travel recommendations.

Relevance: Informed the implementation of our geolocation-based suggestions using Google Places API.

These papers collectively underscore the growing need for smart, context-aware, and personalized travel systems. Our project not only implements the findings from these studies but also goes further by integrating them into a cohesive, full-stack application ready for deployment.

CHAPTER 3

REQUIREMENTS AND ANALYSIS :

3.1 Requirements Specification :

Requirements specification is the foundational step that defines the functional and non-functional expectations of the system. For the Smart Travel Planner and Booking System, the requirements were derived based on real-world scenarios, user interviews, competitor analysis, and research findings. These specifications ensure that the system is designed to meet user demands, industry standards, and future scalability.

◆ **Functional Requirements**

These specify what the system should do:

1. **User Registration and Authentication**
 - Signup with email and password.
 - Login with credential validation.
 - Password recovery via OTP email verification.
2. **Search and Booking Functionality**
 - Search destinations using keywords and filters.
 - Book flights, trains, buses, and accommodations.
 - View and manage bookings via "My Bookings".
3. **Personalized Itinerary Generator**
 - Generate recommended travel plans based on preferences.
 - Save, edit, and finalize itinerary.
4. **Chatbot Assistant**
 - Real-time query resolution.
 - Navigation help and travel recommendations.
5. **Location-Based Recommendations**
 - Suggest nearby attractions and hotels using geolocation.
6. **Admin Panel**
 - Manage users, view reports, handle bookings.
 - Display analytics and booking trends using charts.
7. **Carbon Footprint Estimator & Packing List Generator**
 - Show environmental impact of chosen transport.
 - Suggest items based on weather and destination.

◆ **Non-Functional Requirements**

These define how the system should perform:

1. **Performance:** The system should respond to user actions within 2 seconds.
2. **Security:** Passwords are encrypted; OTP verification and SSL encryption are enforced.
3. **Scalability:** Designed for deployment on scalable cloud platforms (AWS/Heroku).
4. **Usability:** Interface must be intuitive, responsive, and mobile-friendly.
5. **Maintainability:** Follows MVC architecture to ensure modularity and easy updates.
6. **Reliability:** Ensures consistent behavior under valid inputs and handles failures gracefully.

3.2 Planning and Scheduling :

Effective planning and scheduling are critical to ensure the timely and successful completion of the project. This section outlines the phases, timelines, and tasks that structured the development process of the Smart Travel Planner and Booking System.

Phase	Description	Duration
1. Requirement Analysis	Gathered features from users, researched APIs and tools	1 Week
2. System Design	Created flowcharts, ER diagrams, and defined database schema	1 Week
3. Backend Development	Set up Java Servlets, Spring controllers, JDBC integration	2 Weeks
4. Frontend Development	Designed JSP pages, login, booking forms, chatbot UI	2 Weeks
5. API Integration	Integrated travel API, weather API, Google Places, Razorpay	1.5 Weeks
6. Advanced Features	Implemented AI itinerary generation, chatbot, carbon calculator	2 Weeks
7. Testing & Debugging	Unit testing (JUnit), browser automation (Selenium)	1 Week
8. Deployment & Documentation	Deployed on Tomcat/AWS, prepared report & demo	1 Week

Gantt Chart Overview (Described for the Report)

- **Week 1–2:** Requirement Gathering + UI Wireframes
- **Week 3–4:** Core Module Development
- **Week 5:** Integration of APIs and databases
- **Week 6–7:** Testing and debugging
- **Week 8:** Deployment and Final Review

Project management was facilitated using **Trello** for task tracking and **Git** for version control. Weekly review meetings ensured that deliverables aligned with goals.

3.3 Software and Hardware Requirements :

Hardware Requirements

Component	Specification
Server	Intel i5 or higher, 8–16 GB RAM, SSD, 1 Gbps NIC
Client Devices	Desktop/Laptop/Smartphone with modern browser
Storage	Minimum 100 GB SSD on server
Network	Stable internet connection (minimum 10 Mbps)

The application is designed to run on a standard cloud instance or university lab machines without requiring high-end configurations.

Software Requirements

Software Component	Details
OS	Windows 10+ / Ubuntu 20+ / Mac OS
Java	JDK 8 or higher
IDE	IntelliJ IDEA / Eclipse / NetBeans
Application Server	Apache Tomcat 9+ or Spring Boot
Build Tools	Maven or Gradle
Database	MySQL 8.0 / PostgreSQL
Frontend	JSP, HTML5, CSS3, JavaScript, Bootstrap
APIs Used	Amadeus API, Google Places API, OpenWeatherMap, Razorpay
Security Tools	JavaMail for OTP, HTTPS for encryption
Testing	JUnit, Selenium WebDriver
Deployment	AWS EC2 / Heroku / Google Cloud Platform

All selected tools are open-source or free for developers, keeping the system cost-effective and suitable for educational and startup environments.

3.4 Preliminary Product Description :

The **Smart Travel Planner and Booking System** is a full-stack, web-based application designed to simplify and personalize the travel experience for users. It combines traditional travel booking modules with intelligent features to offer an innovative, user-centric platform.

Core Capabilities

- Users can **register and login** to access a personalized dashboard.
- They can **search for destinations**, view transport options (flights, buses, trains), and **book trips**.
- The system integrates third-party APIs to provide **live data on transport and destinations**.
- Users can use the **AI itinerary builder** to generate complete trip plans tailored to their interests and budgets.
- The **chatbot assistant** provides instant help, answers queries, and guides users through bookings.
- **Geo-suggestions** help users find attractions and services near their current or target location.
- **Weather-based packing suggestions** and **carbon footprint estimates** are generated based on chosen travel methods.

Admin Capabilities

- Admins have a dedicated dashboard where they can:
 - View and manage bookings
 - Manage user accounts
 - Monitor traffic and popular destinations
 - Visualize business metrics using graphs and statistics

User Experience

- Mobile-friendly design ensures usability across devices.
- Visual feedback, tooltips, and real-time messages make the interface intuitive.
- Users receive **email confirmations** and can track booking statuses from their profile.

The system is modular, allowing future enhancements like hotel selection, ride-sharing integration, and multilingual support. It combines **scalability**, **security**, and **AI-driven personalization** to deliver a next-gen travel platform.

CHAPTER 4

PROPOSED METHODOLOGY :

4.1 System Architecture and Design Methodology:

The Smart Travel Planner and Booking System is built using a modular, layered architecture based on the Model-View-Controller (MVC) design pattern. This ensures clean separation of concerns, making the system maintainable, scalable, and testable.

Layer	Description
Model	Contains Java classes representing the core entities like User, Booking, Itinerary, and Preference. It also includes business logic and DAO (Data Access Objects) for database communication.
View	Comprises JSP files (e.g., login.jsp, booking.jsp, dashboard.jsp) that generate the user interface with HTML, CSS, and JavaScript.
Controller	Built using Spring MVC. Controllers handle user requests, process input, call the appropriate service logic, and return responses to views.

Request Flow

- User Action: User searches for flights and clicks "Book".
- View Layer: booking.jsp form is submitted.
- Controller: /bookFlight controller method is called.
- Service Layer: Validates input, interacts with API for flight data, calculates pricing.
- Model/DAO: Saves booking in bookings table.
- Response: Confirmation page rendered with payment details.

API Integrations

- Travel APIs (like Amadeus or Skyscanner): Real-time transport options.
- Google Places API: Location-based recommendation engine.
- OpenWeatherMap API: Fetch weather data for packing suggestions.
- Razorpay/Stripe: Secure online payment gateways.
- Email API (JavaMail): For OTP verification and booking confirmations.

Security Layers

- User Input Validation: Server-side checks to prevent SQL injection, XSS, CSRF.
- Password Encryption: Hashing with salted encryption algorithms (e.g., SHA-256).
- OTP Verification: Sent via email for password recovery and sensitive actions.
- Session Management: Prevent unauthorized access after login.

4.2 Feature Development Workflow and AI Enhancements:

To enrich the user experience beyond basic functionality, the system incorporates intelligent and automated features using machine learning logic, RESTful services, and real-time data processing.

AI-Based Itinerary Generator

- Input: User's travel history, preferences (e.g., beaches, budget range, trip duration).
- Logic: A collaborative filtering or content-based filtering model is trained using sample destination metadata and user behavior.
- Output: Suggested destinations, accommodation, and activities tailored to user profile.
- Implementation: Runs as a backend module with optional Python integration via REST API. Frontend fetches results via AJAX.

Chatbot Assistant (NLP-Based)

- Built with either:
 - Dialogflow (Google) or
 - GPT-based OpenAI API
- Function: Answer user questions like “Show me places to visit in Himachal,” or “How to reset my password?”
- Integration: Chat UI embedded in all pages via chatbot.jsp, linked to Spring controller /chatbot-query.

Location-Based Recommendations

- JavaScript in dashboard.jsp fetches user's geolocation.
- AJAX sends location to backend controller.
- Controller calls Google Places API to fetch top-rated attractions.
- Results are dynamically displayed on the home page or search result page.

Smart Packing List Generator

- Takes destination and date range from user.
- Calls OpenWeatherMap API to get forecast.
- Matches forecast with clothing/accessory rules.
 - E.g., Rain = umbrella; Cold = jacket.
- Returns a checklist in packing-list.jsp.

Carbon Footprint Calculator

- Each transport mode has CO₂ emission coefficient (e.g., flight = 0.115 kg/km).
- Calculates distance using Maps API.
- Multiplies by coefficient and displays results post-booking.
- Encourages users to choose eco-friendly options.

Admin Analytics Dashboard

- Uses Chart.js to visualize:
- Number of bookings over time
- Top destinations
- Revenue trends
- /admin/analytics endpoint fetches data as JSON
- Interactive graphs update dynamically

Development Workflow Summary

Step	Action
1. Define Requirements	User stories and functional specs
2. Design UI/UX	Wireframes and navigation flow
3. Set Up Backend	Spring controllers, DAO classes
4. Create Database Schema	Tables for users, bookings, preferences
5. Integrate APIs	Travel, payment, weather, chatbot
6. Test	Unit testing with JUnit, UI with Selenium
7. Deploy	Host on Tomcat/AWS/Heroku

CHAPTER 5

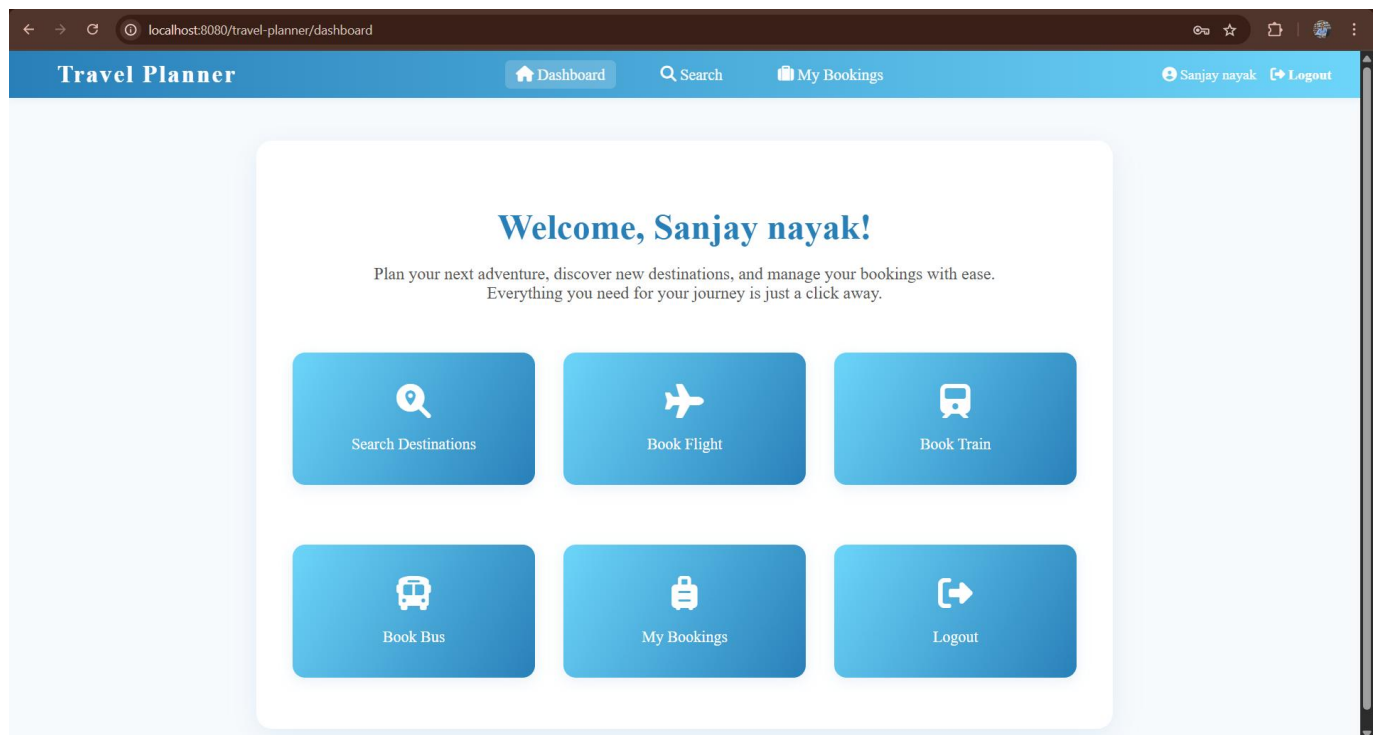
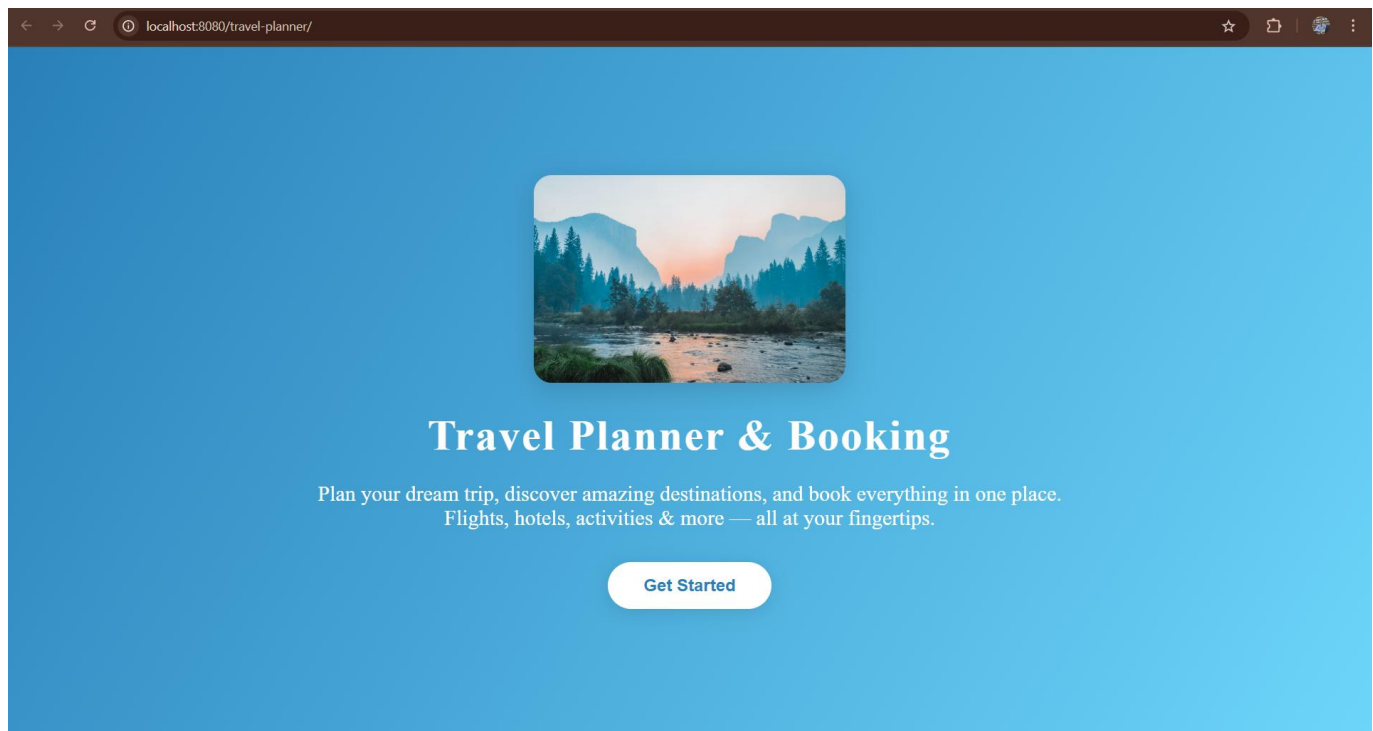
RESULTS:

The Smart Travel Planner and Booking System achieved its intended objectives by delivering a comprehensive, intelligent, and user-friendly platform for end-to-end travel planning and booking. The project successfully integrates advanced Java technologies such as JSP, Servlets, JDBC, and the Spring Framework to create a dynamic and modular backend, while also employing responsive frontend design with HTML, CSS, and JavaScript to ensure an intuitive user experience across devices. Users can register, log in, recover forgotten passwords via OTP, search for destinations, and book flights, trains, and buses seamlessly. Real-time integration with travel APIs enables users to access updated data for transportation and destinations, ensuring relevance and reliability throughout the booking process.

One of the most significant outcomes of this project was the integration of intelligent features that enhanced user engagement and automation. The AI-powered itinerary generator personalizes travel suggestions based on user interests, history, and budget, thereby reducing the effort required in manual planning. The embedded chatbot assistant provides immediate help to users by answering frequently asked questions and guiding them through the platform's functionalities. Additionally, the geolocation-based recommendation feature dynamically suggests nearby tourist attractions and accommodations based on the user's current or target location, adding a context-aware layer to the system.

The system's ability to adapt to real-world conditions was further validated through the weather-based packing list generator, which provides smart suggestions for travel essentials based on destination forecasts. Moreover, the inclusion of a carbon footprint calculator reflects a modern, environmentally conscious approach, helping users make informed, eco-friendly travel decisions. From an administrative perspective, the project delivered a robust and visually informative dashboard where administrators can view user statistics, analyze booking trends, and manage system operations efficiently through interactive charts and metrics.

Through rigorous testing using tools like JUnit and Selenium, the system demonstrated high reliability, minimal error rates, and efficient handling of user requests. The platform is also cloud-deployable and scalable, making it suitable not just for academic demonstration but also for real-world implementation by travel startups or agencies. In conclusion, the Smart Travel Planner and Booking System successfully combines functionality, intelligence, and modern usability into a single, innovative platform that redefines the way users plan and experience travel.



CHAPTER 6

CONCLUSION:

In conclusion, the **Smart Travel Planner and Booking System** successfully addresses the limitations of traditional travel planning by offering a centralized, intelligent, and user-friendly web application. It brings together essential travel services—such as destination discovery, transport booking, and itinerary management—into a single platform powered by advanced Java technologies, real-time APIs, and AI-based automation. The integration of personalized itinerary generation, chatbot assistance, location-based suggestions, weather-aware packing lists, and carbon footprint estimation significantly enhances the user experience while promoting smart and sustainable travel practices.

The project not only meets the functional goals of simplifying travel planning but also demonstrates scalability, modular design, and real-world applicability. Its robust architecture, secure user management, and intuitive interface ensure that it can be used confidently by both individual travelers and administrative personnel. Overall, this system represents a forward-thinking solution that blends technical proficiency with practical usability, making it a valuable contribution to the domain of intelligent travel systems and a strong foundation for future enhancements and real-world deployment.

CHAPTER 7

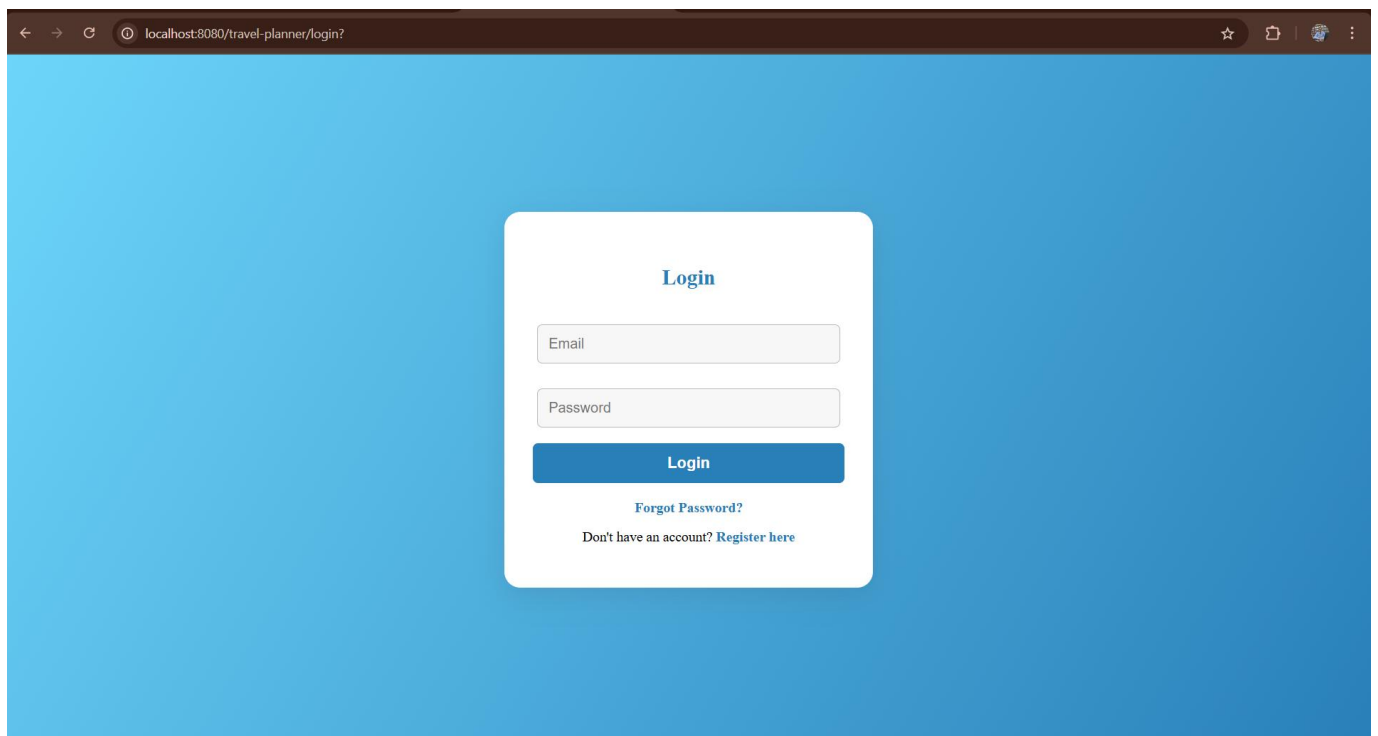
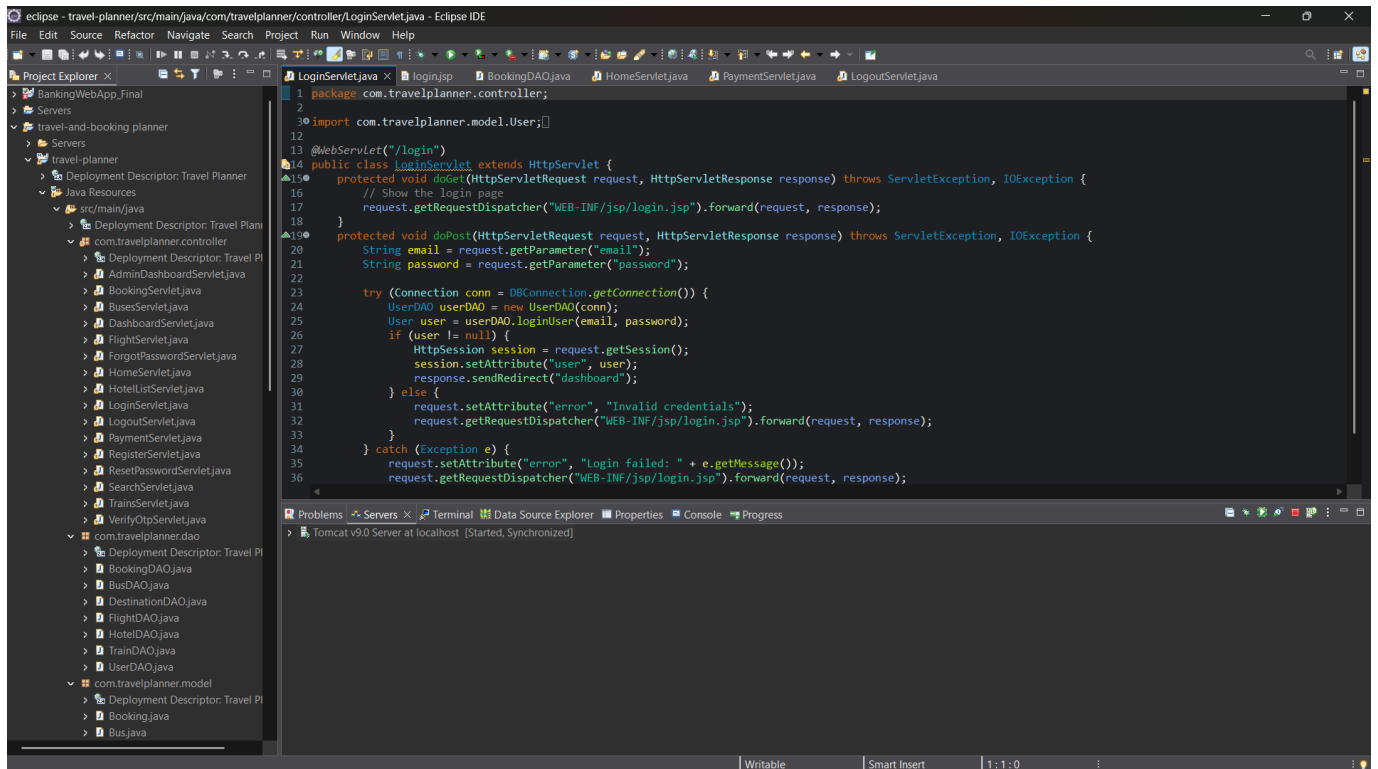
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CHAPTER 8

APPENDICES



```

1  <%@ page contentType="text/html; charset=UTF-8" %>
2
3  <!DOCTYPE html>
4  <html lang="en">
5  <head>
6
7      <title>Login - Travel Planner</title>
8      <link rel="stylesheet" href="../static/css/style.css">
9      <style>
10         body {
11             background: linear-gradient(120deg, #6dd5fa 0%, #2980b9 100%);
12             min-height: 100vh;
13             margin: 0;
14             display: flex;
15             align-items: center;
16             justify-content: center;
17         }
18         .auth-card {
19             background: #fff;
20             padding: 2.5em 2em 2em 2em;
21             border-radius: 18px;
22             box-shadow: 0 8px 32px rgba(44, 62, 80, 0.15);
23             width: 350px;
24             max-width: 95vw;
25             text-align: center;
26         }
27         .auth-card h2 {
28             color: #2980b9;
29             margin-bottom: 1.2em;
30             font-weight: 700;
31         }
32         .auth-card input[type="email"],
33         .auth-card input[type="password"] {
34             width: 90%;
35             padding: 12px;
36             margin: 10px 0 18px 0;
37             border: 1px solid #2b2b2b;
38             border-radius: 6px;
39             font-size: 1em;
40             background: #fff;
41         }
42         .auth-card input[type="submit"] {
43             width: 100%;
44             background: #2980b9;
45             color: #fff;
46             border: none;

```

```

1  package com.travelplanner.controller;
2
3  import com.travelplanner.dao.DestinationDAO;
4
5  @WebServlet("/search")
6  public class SearchServlet extends HttpServlet {
7
8      protected void doGet(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {
9          String keyword = request.getParameter("keyword");
10         if (keyword == null) keyword = "";
11         try (Connection conn = DBConnection.getConnection()) {
12             DestinationDAO dao = new DestinationDAO(conn);
13             List<Destination> results = dao.searchDestinations(keyword);
14             request.setAttribute("results", results);
15             request.getRequestDispatcher("/WEB-INF/jsp/search.jsp").forward(request, response);
16         } catch (Exception e) {
17             request.setAttribute("error", "Search failed: " + e.getMessage());
18             request.getRequestDispatcher("/WEB-INF/jsp/search.jsp").forward(request, response);
19         }
20     }
21 }

```

```
eclipse - travel-planner/src/main/webapp/WEB-INF/jsp/search.jsp - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help

Project Explorer
  Deployment Descriptor: Travel P
  DBConnection.java
  MySQLCleanupContextListener.js
  com.travelplanner.util
    Deployment Descriptor: Travel P
    EmailUtil.java
  src/main/resources
  src/test/java
  src/test/resources
  Libraries
  Referenced Libraries
  Deployed Resources
  db
  src
    main
      java
      resources
      webapp
        static
          WEB-INF
            jsp
              admin_dashboard.jsp
              bookings.jsp
              buses.jsp
              dashboard.jsp
              flights.jsp
              hotels.jsp
              login.jsp
              payments.jsp
              register.jsp
              search.jsp
              trains.jsp
              web.xml
              forgot-password.jsp
              index.jsp
              reset-password.jsp
              verify-otp.jsp
    test
    target
    pom.xml

1  <%@ page contentType="text/html; charset=UTF-8" %>
2
3
4
5  <%@ page import="java.util.List" %>
6  <%@ page import="com.travelplanner.model.Destination" %>
7  <%@ page session="true" %>
8  <%
9      Object userObj = session.getAttribute("user");
10     if (userObj == null) {
11         response.sendRedirect("${pageContext.request.contextPath}/login");
12         return;
13     }
14     List<Destination> results = (List<Destination>) request.getAttribute("results");
15 %>
16 <!DOCTYPE html>
17 <html lang="en">
18 <head>
19     <title>Search Destinations - Travel Planner</title>
20     <link rel="stylesheet" href="../../static/css/style.css">
21     <style>
22     <body {
23         background: #f7fafd;
24         margin: 0;
25         padding: 0;
26     }
27     .navbar {
28         width: 100%;
29         background: linear-gradient(90deg, #2980b9 0%, #6dd5fa 100%);
30         color: #fff;
31         padding: 0.7em 0;
32         box-shadow: 0 2px 8px rgba(41,128,185,0.08);
33         display: flex;
34         align-items: center;
35         justify-content: space-between;
36         position: sticky;
37         top: 0;
38         z-index: 100;
39     }
40     .navbar .logo {
41         font-size: 1.7em;
42         font-weight: bold;
43         margin-left: 2em;
44         letter-spacing: 2px;
45     }
46     .navbar ul {
```

